The Tung Tulip Circles of a Quadrangle

A proposed new item for the *Encyclopedia of Quadri-Figures*: two circles obtained by inverting the Gergonne–Steiner point in the nine-point circles and the Euler–Poncelet point in the midray circles

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September 9, 2026

Abstract

Let $P_1P_2P_3P_4$ be a quadrangle, N_i the nine-point circle of the component triangle $P_jP_kP_l$, and M_i the *midray circle* of P_i (the circle through the midpoints of P_iP_j, P_iP_k, P_iP_l). In the notation of the Encyclopedia of Quadri-Figures (EQF), the four circles N_i concur at the Euler–Poncelet point QA-P2 and the four circles M_i concur at the Gergonne–Steiner point QA-P3. We show that the four inverses of QA-P3 in the circles N_i lie on a circle Γ_1 through QA-P2, and the four inverses of QA-P2 in the circles M_i lie on a circle Γ_2 through QA-P3; the two circles are reflections of each other in the centroid QA-P1. We call them the *1st and 2nd Tung Tulip circles* and their centers the *Tung Tulip points*. The proof is a one-line inversion argument that identifies Γ_1 with the Steiner line of a complete quadrilateral whose Miquel point is the image of QA-P3. For a cyclic quadrangle on the unit circle the centers are $(\sigma_1\sigma_2-\sigma_3)/(2\sigma_2)$ and $\sigma_3/(2\sigma_2)$ and the common radius is $|\sigma_1|/(2|\sigma_2|)$. As a by-product we note that QA-P23 is the centroid of the inverses of QA-P4 in the four circumcircles.

1 Notation

$P_1P_2P_3P_4$ is the reference quadrangle; $\{i,j,k,l\} = \{1,2,3,4\}$. We use the EQF names (`https://www.chrisvantienhoven.nl/epg/n-geometry/quadri-figures/qa-items/`):

- QA-P1: the centroid $G = \frac{1}{4}(P_1+P_2+P_3+P_4)$;
- QA-P2: the Euler–Poncelet point, the common point of the four nine-point circles N_i of the component triangles $P_jP_kP_l$;
- QA-P3: the Gergonne–Steiner point, the common point of the four midray circles M_i, where M_i passes through the midpoints of P_iP_j, P_iP_k, P_iP_l. QA-P3 is the reflection of QA-P2 in QA-P1.

O_i, H_i, R_i denote circumcenter, orthocenter and circumradius of $P_jP_kP_l$; $\mathrm{mid}(XY)$ the midpoint of XY. "Inverse of a point X in a circle C" is the image of X under the inversion in C.

2 Definition and main theorem

Definition 1 (Tung Tulip circles and points)**.** Let X_i be the inverse of QA-P3 in the nine-point circle N_i, and Y_i the inverse of QA-P2 in the midray circle M_i $(i = 1,2,3,4)$.

- The *1st Tung Tulip circle* Γ_1 is the circle through X_1, X_2, X_3, X_4; its center T_1 is the *1st Tung Tulip point*.

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†Tung Tulip is an *AI persona* of the first author: the name under which the first author conducts research together with AI systems such as Claude, Codex and Gemini. The results in this note were found, proved and verified through this collaboration; the mathematical responsibility rests with the first author.

- The *2nd Tung Tulip circle* Γ_2 is the circle through Y_1, Y_2, Y_3, Y_4; its center T_2 is the *2nd Tung Tulip point*.

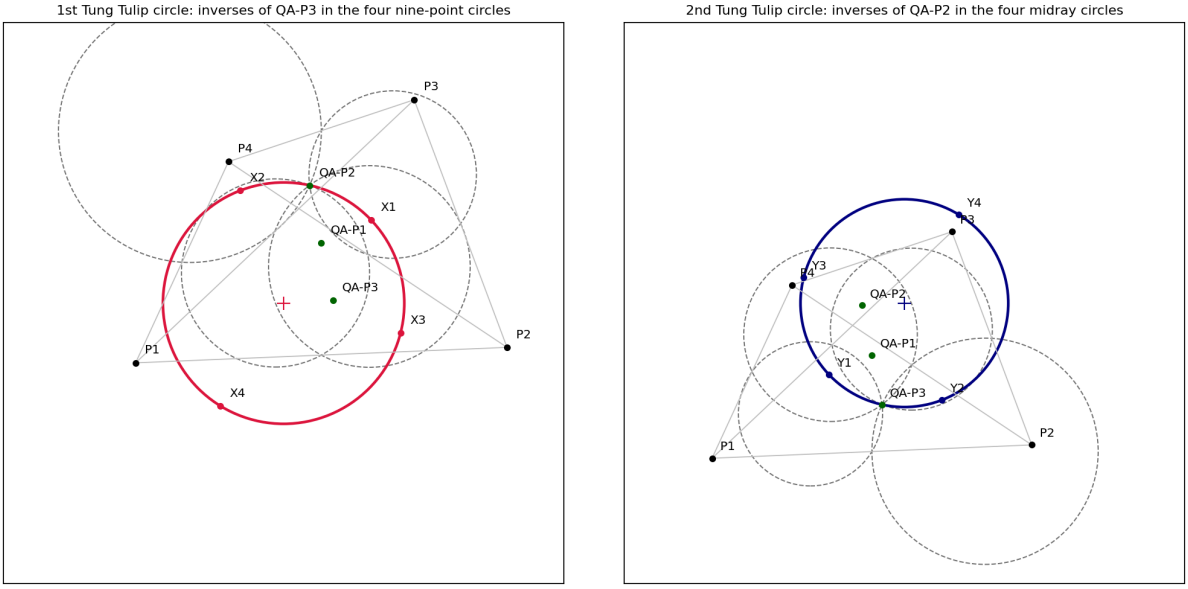


Figure 1: Left: the four nine-point circles (dashed), QA-P2, QA-P3, the inverses X_i and the 1st Tung Tulip circle Γ_1 (through QA-P2). Right: the four midray circles, the inverses Y_i and the 2nd Tung Tulip circle Γ_2 (through QA-P3). The two circles are symmetric about QA-P1.

Theorem 1. *For every quadrangle (no three vertices collinear, QA-P2 $\neq$ QA-P3):*

(a) X_1, X_2, X_3, X_4 *are concyclic, and their circle* Γ_1 *passes through* QA-P2*;*

(b) Y_1, Y_2, Y_3, Y_4 *are concyclic, and their circle* Γ_2 *passes through* QA-P3*;*

(c) Γ_2 *is the reflection of* Γ_1 *in* QA-P1*; in particular* Γ_1, Γ_2 *have the same radius and* QA-P1 *is the midpoint of* T_1T_2*.*

Lemma 2. *The reflection in* QA-P1 *maps the nine-point circle* N_i *onto the midray circle* M_i*, for each* i*.*

Proof. Both circles have radius $R_i/2$: N_i is the image of the circumcircle of $P_jP_kP_l$ under the homothety $(H_i, \frac{1}{2})$, and M_i is its image under the homothety $(P_i, \frac{1}{2})$. Their centers are $\frac{1}{2}(O_i + H_i)$ and $\frac{1}{2}(P_i + O_i)$. With vectors from an arbitrary origin, $H_i = P_j + P_k + P_l - 2O_i$, hence $\frac{1}{2}(O_i + H_i) + \frac{1}{2}(P_i + O_i) = \frac{1}{2}(P_1 + P_2 + P_3 + P_4) = 2\,$QA-P1, i.e. the two centers are symmetric about QA-P1. □

Proof of Theorem 1. (a) Fix a circle centered at QA-P2 and let ι be the inversion in it. Since every N_i passes through QA-P2, the images $\ell_i = \iota(N_i)$ are four lines; they form a complete quadrilateral $\mathcal{L}$. Two nine-point circles N_j, N_k meet at QA-P2 and at the midpoint of the common side P_lP_m of their triangles, so $\ell_j \cap \ell_k = \iota(\mathrm{mid}(P_lP_m))$. Consequently the triangle bounded by ℓ_j, ℓ_k, ℓ_l has vertices $\iota(\mathrm{mid}(P_iP_j)), \iota(\mathrm{mid}(P_iP_k)), \iota(\mathrm{mid}(P_iP_l))$, and its circumcircle is $\iota(M_i)$. The four circles M_i concur at QA-P3, so the four circumcircles of the component triangles of $\mathcal{L}$ concur at ι(QA-P3): *the Miquel point of* $\mathcal{L}$ *is* ι*(QA-P3)*.

By Steiner's theorem the reflections of the Miquel point in the four lines of $\mathcal{L}$ lie on a line s (the Steiner line, which also carries the four orthocenters). Conjugating by ι, the reflection in ℓ_i becomes the inversion in N_i, so ι maps the reflection of ι(QA-P3) in ℓ_i to the inverse X_i of QA-P3 in N_i. Hence $X_1, \ldots, X_4$ lie on $\iota(s)$, which is a circle through the center QA-P2 of ι (a line: only in the degenerate case $s \ni$ QA-P2).

(b) is the same argument with the roles of QA-P2/QA-P3 and N_i/M_i exchanged: $M_j \cap M_k = \{\text{QA-P3}, \mathrm{mid}(P_jP_k)\}$, the triangle bounded by $\iota'(M_j), \iota'(M_k), \iota'(M_l)$ (ι' an inversion centered at QA-P3) has vertices the images of the midpoints of the sides of $P_jP_kP_l$, its circumcircle is $\iota'(N_i)$, and the concurrence of the N_i at QA-P2 makes $\iota'(\text{QA-P2})$ the Miquel point.

(c) The reflection in QA-P1 swaps QA-P2 and QA-P3 and, by Lemma 2, swaps N_i and M_i; it therefore swaps X_i and Y_i, and Γ_1 with Γ_2. □

3 Properties

P1. $\Gamma_1 \ni$ QA-P2, $\Gamma_2 \ni$ QA-P3, $\Gamma_2 =$ reflection of Γ_1 in QA-P1, $T_2 = 2\,\text{QA-P1} - T_1$.

P2. (Inversive description.) For any inversion ι centered at QA-P2, $\iota(\Gamma_1)$ is the Steiner line of the quadrilateral formed by the inverses of the four nine-point circles, whose Miquel point is $\iota(\text{QA-P3})$. Dually for Γ_2 with an inversion centered at QA-P3.

P3. (Tangent pedal circle.) For each i let F_i be the second intersection of N_i with the circle through QA-P2 and QA-P3 orthogonal to N_i. The points $F_1, \ldots, F_4$ lie on a circle through QA-P2 which is *tangent to* Γ_1 *at* QA-P2. (It is the image of the pedal line QL-L3 of the Miquel point, which is parallel to the Steiner line.)

P4. The inverse of Γ_1 in each nine-point circle N_i is a circle through QA-P3; the inverse of Γ_2 in each midray circle M_i is a circle through QA-P2.

P5. Lemma 2 contains the EQF statement "QA-P3 is the reflection of QA-P2 in QA-P1" as a corollary.

P6. (Cyclic quadrangle.) If $P_1, \ldots, P_4$ lie on the unit circle with $\sigma_1 = \sum a$, $\sigma_2 = \sum ab$, $\sigma_3 = \sum abc$, $\sigma_4 = abcd$ (so that QA-P2 $= \sigma_1/2$ and QA-P3 $= O = 0$), then

$$T_1 = \frac{\sigma_1\sigma_2 - \sigma_3}{2\sigma_2} = \frac{\sigma_1}{2} - \frac{\sigma_3}{2\sigma_2}, \qquad T_2 = \frac{\sigma_3}{2\sigma_2} = \tfrac{1}{2}\,\overline{\left(\frac{\sigma_1}{\sigma_2}\right)}, \qquad r(\Gamma_1) = r(\Gamma_2) = \frac{|\sigma_1|}{2|\sigma_2|}.$$

In particular Γ_2 passes through the circumcenter O and its antipode on Γ_2 is σ_3/σ_2; T_2 is not in the $\mathbb{Q}$-span of σ_1 and $P + E + F$ (the vertices of the diagonal triangle), unlike all EQF points QA-P1…P13, P20…P43 computed for the cyclic case.

Remark 1 (A by-product)**.** The same experiments produced a new property of an existing item: *QA-P23 (the Inscribed Square Axes Crosspoint) is the centroid of the four inverses of QA-P4 (the Isogonal Center) in the four circumcircles of the component triangles.* Since $|\text{QA-P4}\,P_i| \cdot R_i$ is constant, these inverses lie on the lines QA-P4 O_i.

4 Coordinates

All objects are rational in the vertices. In the cyclic case the coordinates are the closed forms of P6. For a general quadrangle the CT-coordinates of T_1 are rational functions of high degree in (a^2, b^2, c^2, p, q, r); a symbolic computation is included in the accompanying files (`research/tt_symbolic_ct.py`); the construction of P2 (inverse, about QA-P2, of the reflection of QA-P2 in the Steiner line of $\mathcal{L}$) gives T_1 with ruler and compass.

5 Verification

Every statement above was checked numerically on 200 random quadrangles and 200 random cyclic quadrangles (`verify/verify_tung_tulip.py`); the largest relative discrepancy is below 10^{-7} for all items (below 10^{-9} for the concyclicities). The search that led to these circles used a local knowledge

base of all 111 EQF QA-items (`eqf-qa-knowledge/`): none of them mentions inversion in the nine-point or midray circles, and, among the 36 catalogued QA-points available in our registry (QA-P1–P13, P16–P22, P25–P27, P29–P34, P36, P37, P39, P43), the circles Γ_1, Γ_2 pass only through QA-P2 resp. QA-P3, while T_1, T_2 lie on none of QA-L1–L10 and on no line joining two of these points.

References

[1] C. van Tienhoven, *Encyclopedia of Quadri-Figures*, `https://www.chrisvantienhoven.nl/epg/n-geometry/quadri-figures/`, items QA-P1, QA-P2, QA-P3, QA-P4, QA-P23, QL-P1, QL-L2, QL-L3.

[2] J. Steiner, *Théorèmes sur le quadrilatère complet*, Annales de Gergonne 18 (1827–28); the collinearity of the reflections of the Miquel point (and of the four orthocenters) is the "Steiner line" of a complete quadrilateral.